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Search for Type III Seesaw Model Heavy Fermions in Events with Four Charged Leptons using 5.8 fb^{-1} of $\sqrt{s}$ = 8 TeV data with the ATLAS Detector

The ATLAS Collaboration

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Abstract

A search for Type III Seesaw model heavy fermions is presented. The search is performed in a data sample corresponding to 5.8 fb^{-1} of integrated luminosity collected in 2012 in pp collisions at $\sqrt{s}$ = 8 TeV by the ATLAS detector at the LHC. Charged heavy fermions, $N^{\pm}$, are reconstructed in the channel $Z(\ell\ell)+\ell^{\pm}$ together with an additional charged lepton from the decay of a partner heavy fermion, $N^{0/\mp} \to \ell X$, where $\ell = e, \mu$. No evidence of the $N^{\pm}$ is observed. Upper limits at 95% confidence level are set on the production cross section of $N^{+}N^{0}$ times the branching fraction to the examined four lepton final states. When interpreted in the context of the Type III Seesaw model, the results translate into a lower limit on the mass of the N states of 245 GeV at 95% confidence level.

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1 Introduction

In the Standard Model (SM) neutrinos are massless; this is in contradiction with the evidence from neutrino oscillation experiments, showing that at least two neutrinos of all three generations are massive [1, 2]. While it is possible to measure the mass difference between neutrino flavours through neutrino oscillations, the absolute neutrino mass scale remains unknown. Analysis of the cosmic microwave background places the most stringent constraint on the neutrino mass scales, setting an upper limit of 0.3 eV on the sum of all neutrino masses [3]. A number of theoretical models have been put forward to augment the SM to explain not only how neutrinos acquire mass, but also why their mass is so small [3]. So far, there is almost no experimental evidence to refute or confirm many of these models.

One theoretically well motivated approach to neutrino mass generation is the addition of Majorana and possibly Dirac mass terms to the SM Lagrangian [4, 5, 6, 7]. The way in which these mass terms are added is not unique. However, three possible mechanisms enter at tree level: Seesaw models Type I [8, 9, 10], II [11, 12, 13, 14, 15] and III [16, 17, 18]. In all three cases, the mass scale of any new particle added to the SM is typically large, suppressing the mass of the observed light neutrinos, through seesaw like mechanisms. The Type I Seesaw mechanism predicts at least two right-handed neutrinos, ν_R which must have masses of the order of 10^{14} GeV to account for neutrino masses of the order of eV, far beyond the direct reach of any present, or future collider experiment [5]. In the Type II Seesaw mechanism, neutrino masses are generated in the Higgs sector, by adding an additional Higgs triplet [4]. The Type III Seesaw mechanism introduces at least two fermionic triplets [16] that generate neutrino masses. The lightest fermionic triplet has two charged heavy fermions denoted N^+ and N^-, and a neutral heavy fermion N^0, generating a reciprocal seesaw mechanism. The masses of these states are approximately degenerate.

Type II and III Seesaw heavy states couple to gauge bosons, hence the possible smallness of the Yukawa coupling does not affect the production cross section of such heavy states at the LHC. Furthermore, the mass of the heavy fermion (m_N) may be of order 100 GeV, when either the Yukawa couplings are very small or if an inverse seesaw were at work [19]. At the LHC, the Type III Seesaw model manifests itself through pair production of heavy fermions, $N^0N^\pm$ and $N^\pm N^\mp$, where the production cross section for $N^0N^\pm$ is approximately twice that of $N^\pm N^\mp$. The N^0 and the $N^\pm$ fermions decay into a W, Z or H and an accompanying charged or neutral lepton. This paper considers events with four or more leptons (electrons or muons)[1], allowing for the explicit reconstruction of $N^\pm$. A direct search for the $N^\pm$ resonance is performed by fully reconstructing the mode $N^\pm \to Z\ell$, where the on-shell Z decays leptonically and $\ell = e, \mu$. The $N^\pm$ resonance is predicted to be narrow, with a width of less than 1 GeV. The presence of an additional charged lepton from heavy fermion pairs is exploited to further suppress background contributions, leaving the Z boson pair production the dominant one. Four charged lepton final states from heavy fermion pair production, where two leptons are from a Z boson decay, are dominated by $N^\pm \to Z\ell^\pm$ and $N^0 \to W^\pm\ell^\mp$ decays: it comprises 70%-90% of the total rate, depending on the mass of the heavy fermion (based on Ref. [19]). The remainder arises from $N^\pm N^\mp$ production. The study of four-lepton production presented in this paper is interpreted in the context of $N^0N^\pm$ production.

The total width and decay branching fractions to SM particles of the heavy fermions depend on the mixing angles between the SM leptons and the new heavy states $|V_e|$, $|V_\mu|$ and $|V_\tau|$. In the Type III Seesaw model considered here, the maximal allowed mixing angles $|V_e| = 5.5\times10^{-2}$ and $|V_\mu| = 6.3\times10^{-2}$ [20, 21, 22] are used; the mixing angle for tau leptons is taken to be zero ($|V_\tau| = 0$). The $N^0N^\pm$ and $N^\pm N^\mp$ production cross sections do not depend on these mixing parameters, which enter only in the expressions for the N decay widths. Thus the limits produced within this analysis may be interpreted in terms of a range of mixing angles and Yukawa couplings. The model dependent branching fractions of the heavy fermions have been expressed analytically as a function of the couplings in Ref. [19].

[1] In this paper lepton refers to e or μ.

This is the first search for Type III Seesaw fermions with the ATLAS experiment. Previous searches for seesaw fermions at the LHC were performed by CMS in $\sqrt{s}$ = 7 TeV data. They placed limits on the mass of the heavy fermions by studying three-lepton final states in modes with electrons, muons and missing energy [23]. This final state is most sensitive to $N^{\pm} \to W^{\pm}\nu$ and $N^{0} \to W^{\pm}\ell^{\mp}$ decays, which have the largest respective heavy fermion branching fractions. Their limit on the mass of the heavy fermion ranges from 180 to 210 GeV depending on the coupling to tau leptons. In the analysis presented in this note, the four fermion final state was chosen, to fully reconstruct the heavy fermion masses and achieve a better background suppression of all processes other than pair production of Z bosons.

2 Data and Monte Carlo Samples

The dataset used for this analysis was collected in the first half of 2012 using the ATLAS detector [24] at the LHC, corresponding to 5.8 fb^{-1} of integrated luminosity at $\sqrt{s}$ = 8 TeV . The ATLAS apparatus consists of an inner tracking detector surrounded by a thin superconducting solenoid, electromagnetic and hadronic calorimeters, and an external muon spectrometer incorporating large superconducting air-core toroid magnets. The combination of these systems provides charged particle measurements together with efficient and precise lepton measurements over the pseudorapidity range $|\eta| < 2.5$.[2]

Data were collected using inclusive single-muon and single-electron triggers. The two main triggers require isolated leptons with transverse momentum, p_{T}, to be greater than 24 GeV. The isolation criteria is defined as the scalar sum of the p_{T} of other charged particles within $\Delta R = \sqrt{\Delta\phi^2 + \Delta\eta^2} = 0.2$ of the lepton direction (excluding the lepton itself) divided by the lepton p_{T} to be less than 0.12. To recover potential efficiency loss at high p_{T}, a 60 GeV (36 GeV) threshold trigger without isolation is employed in tandem for electrons (muons). The trigger efficiencies are measured as a function of p_{T}, η, and data-taking period using Z boson events. The efficiencies above p_{T} = 25 GeV are approximately 90% for electrons, and 90% (70%) for muons in the barrel (end cap).

For most processes considered in this analysis, separate Monte Carlo (MC) generators are used to produce the hard scattering process and to model the parton showering, hadronisation, and the underlying event. PYTHIA [25] or PYTHIA8 [26] are used for the latter three steps for the signal and some of the background processes. HERWIG [27] is used for the hadronisation and parton showering, and the underlying event is modelled using JIMMY [28]. ALPGEN [29], interfaced to HERWIG, with the MLM [30] matching scheme is used to model the W+jets, Z/γ^*+jets and $W\gamma$ productions. The $W\gamma^*$, $WZ^{(*)}$, VVV and $t\bar{t}+W/Z$ processes were generated using MADGRAPH [31] and PYTHIA for the hadronisation. The cross sections for the $W\gamma$ and $W\gamma^*/WZ^{(*)}$ processes are normalised to the MCFM [32] NLO predictions. These normalisations (K-factors) are from [33]. MC@NLO [34] is used to model $t\bar{t}$ production and $qq \to WW$ processes, using HERWIG for the parton shower and hadronisation and JIMMY [28] for the underlying event modelling. AcerMC [35] is used to model single top-quark production. The production of $gg \to ZZ$ and $gg \to WW$ are simulated with GG2ZZ [36] and GG2WW [37] respectively; $qq \to ZZ$ is simulated with POWHEG [38, 39]. The CT10 parton distribution function (PDF) set [28] is used for the POWHEG and MC@NLO samples, and CTEQ6L1 [40] is used for the ALPGEN, MadGraph, GG2ZZ, GG2WW and PYTHIA8 samples. The MC generators used to model the signal and background processes are summarised in Table 1.

For the signal process, Feynman rules and couplings of the Type III Seesaw model are first determined from its Lagrangian (Ref. [19]) using FeynRules [41]. The decay widths of the new heavy fermions are calculated from the Feynman rules using the BRIDGE program [42], which depends on various model parameters such as mixing angles and couplings. The results are used in the calculation

[2] ATLAS uses a right-handed coordinate system with its origin at the nominal interaction point (IP) in the centre of the detector, and the z-axis along the beam line. The x-axis points from the IP to the centre of the LHC ring, and the y-axis points upwards. Cylindrical coordinates (r, ϕ) are used in the transverse plane, ϕ being the azimuthal angle around the beam line.

Table 1: The MC generators used to model the signal and background processes.

Process	Generator
$qq \to ZZ$	POWHEG [38, 39] + PYTHIA8 [26]
$gg \to ZZ$	GG2ZZ [36]+ PYTHIA [25]
$qq \to WW$	MC@NLO [34]+ PYTHIA
$gg \to WW$	GG2WW [37] + PYTHIA
inclusive Z/γ^*	ALPGEN [29]+HERWIG [27]
inclusive W	ALPGEN [29]+HERWIG
$W\gamma$	ALPGEN [29]+HERWIG
$W\gamma^*, WZ^{(*)}$	MADGRAPH [31]+PYTHIA
VVV	MADGRAPH [31]+PYTHIA
$t\bar{t}$	MC@NLO [34]+HERWIG
$t\bar{t} + W/Z$	MADGRAPH [31]+PYTHIA
tqb	AcerMC [35]+ PYTHIA
tW/tb	MC@NLO [34]+ HERWIG

of matrix elements. MadGraph5 [31] is used to calculate the matrix elements for each process, while MadEvent [43] simulates the initial scattering and decay of the new heavy fermions. PYTHIA8 is used to simulate the underlying physics by adding initial/final state radiation (ISR/FSR), and decays of the Z and W bosons to the events simulated in MadEvent.

The detector acceptance, and reconstruction efficiencies (with respect to the acceptance) of background processes are obtained from a full simulation [44] of the ATLAS detector using GEANT4 [45], and for signal processes a fast simulation is used, based on the parameterisation of the performance of the ATLAS electromagnetic and hadronic calorimeters [46]. Since the data are affected by the detector response to multiple interactions (pile-up) occurring in the same or nearby bunch crossings, the simulated events are re-weighted so that the distribution of the number of minimum-bias interactions per bunch crossing agrees with data. The description of the ATLAS detector responses and the modelling of various physics processes are not perfect and this leads to small differences between simulated events and real data events. These mis-modellings, e.g. pile-up conditions, trigger efficiencies, object identification efficiencies, energy scales and resolutions, and the corresponding corrections, are applied to the simulated samples used in this analysis.

3 Event Selection

Selected events are required to have a primary vertex consistent with the beam spot position and associated to at least three tracks with $p_{\mathrm{T}} > 400$ MeV. In the case of multiple reconstructed vertices, the primary vertex is chosen as the reconstructed vertex with the highest p_{T}^2 of associated tracks. Data quality criteria are applied to events in order to suppress non-collision backgrounds such as cosmic-ray muons, beam related backgrounds, or noise in the calorimeters.

Events containing heavy fermion candidates are pre-selected by requiring at least four leptons (electrons or muons), with p_{T} thresholds of 25 GeV and 10 GeV, for the leading and the three sub-leading leptons, respectively. At least one of the selected leptons is required to match a triggering object. Electron candidates are selected by applying a set of tight identification criteria using a combination of tracking and calorimetric information [47]. They must lie within $|\eta| < 2.47$ with the region $1.37 < |\eta| < 1.52$, corresponding to the boundary between barrel and end-cap calorimeters, excluded. Muon candidates are

Table 2: Cut flow table with the number of expected events for the signal with a mass of m_N = 120 GeV, SM backgrounds, and data. The uncertainties shown are the statistical ones of the MC samples. In this table, $t\bar{t}$ and $t\bar{t}V$ events are considered together. Processes with expected number of events less than 0.1 are neglected. The last two rows are a decomposition of the third row.

	Signal [120 GeV]	ZZ	Z+jets	VVV	$t\bar{t}V$	Total Bkg.	Data
Four or more leptons	115.0 ± 3.0	61.9 ± 0.3	$1.3^{+2.1}_{-0.8}$	0.53 ± 0.01	1.05 ± 0.11	$64.9^{+2.1}_{-0.8}$	58
Z reconstruction	103.4 ± 2.8	57.1 ± 0.3	$1.3^{+2.1}_{-0.8}$	0.45 ± 0.01	0.99 ± 0.04	$59.9^{+2.1}_{-0.8}$	55
Second Z Veto	88.6 ± 2.6	17.2 ± 0.2	$1.3^{+2.1}_{-0.8}$	0.38 ± 0.01	0.85 ± 0.04	$19.8^{+2.1}_{-0.8}$	19
Third μ	50.5 ± 2.0	9.2 ± 0.1	$1.3^{+2.1}_{-0.8}$	0.20 ± 0.01	0.46 ± 0.03	$11.2^{+2.1}_{-0.8}$	8
Third e	38.1 ± 1.7	8.0 ± 0.1	–	0.18 ± 0.01	0.40 ± 0.02	8.6 ± 0.1	11

formed from tracks measured in the inner detector combined with tracks in the muon spectrometer [48]. They are required to satisfy $|\eta| < 2.5$. Muon quality criteria based on e.g. inner detector hit requirements are applied in order to achieve a precise measurement of the muon momentum and reduce the misidentification rate, particularly against pion/kaon decays-in-flight. Leptons from heavy flavour decays and jets satisfying the lepton identification criteria are suppressed by requiring all four leptons to be isolated. The isolation for the non-triggering leptons are required to be less than 0.15, for muons and 0.16 for electrons.

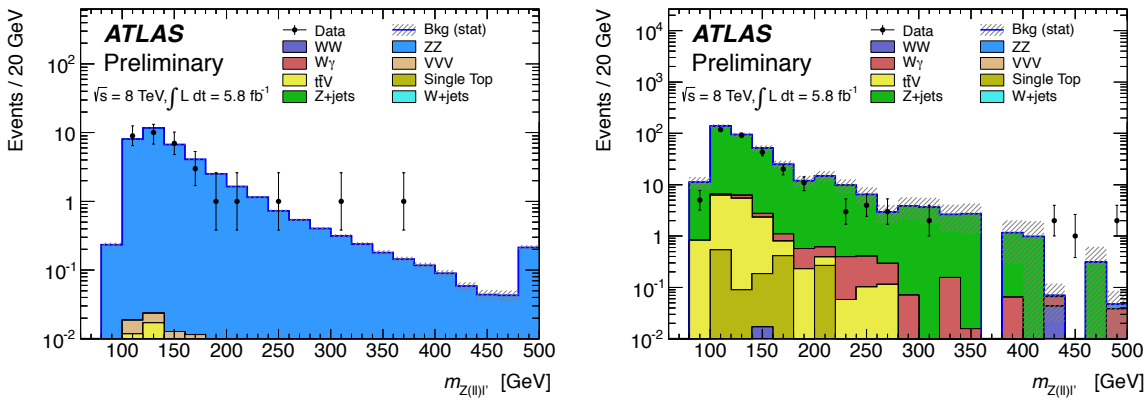
The Z candidates are reconstructed from pairs of opposite sign same flavour leptons, that form an invariant mass within ±10 GeV of the Z mass. The third lepton candidate is identified as the one closest in ϕ to the reconstructed Z. The fourth lepton is the highest p_{T} electron or muon candidate remaining in the event. Events with a second Z boson candidate are rejected. This Z veto reduces contamination from ZZ events. The Z+jets and $t\bar{t}$ background contributions are further reduced by applying impact parameter requirements on the leptons. The impact parameter significance, defined as the absolute value of the impact parameter divided by its uncertainty, $|d_0|/\sigma_{d_0}$, for all leptons is required to be lower than 3.0. The expected yields of signal and background events at several stages of the selection are presented in Table 2. The invariant mass of the $N^{\pm}$, $m_{Z(\ell\ell)\ell'}$, is reconstructed from the kinematics of the Z and the third lepton candidate. The presence of a fourth isolated lepton strongly suppresses problematic Z+jets events and background processes such as WZ, leading to optimal expected signal significance in this data sample.

Background normalisation and control samples

The SM processes that predominantly populate the four lepton signal region are $ZZ^{(*)}$ production, decaying to leptons. These backgrounds are suppressed by vetoing a second Z boson, however there remains a small and irreducible contribution from ZZ^*. The expected yield of background contributions, given in Table 2, shows that although the dominant source of background is ZZ, processes of the type ZZZ^*, ZWW^*, $t\bar{t}V$ and Z+jets can also contribute. Small contributions from $W\gamma$ and top events are also expected and the background from W+jets events is found to be negligible.

The normalisation of the ZZ background is taken from the MC after confirming agreement between the yields of data and MC in a ZZ enhanced control region. The control region is defined by the same selection criteria as the signal region, except for the Z veto, where instead the third and fourth leptons are required to have opposite charge and same flavour, and to form a second Z candidate with an invariant mass within ± 10 GeV of the Z boson. Figure 1(a) shows the distributions of the $m_{Z(\ell\ell)\ell'}$ invariant mass within the control region. The MC agrees with the data within the uncertainty throughout $m_{Z(\ell\ell)\ell'}$, suggesting that the kinematics and rate of the ZZ production is well modelled by the simulation. The ZZ background shape in the signal region is taken from MC.

Processes of the type Z+$b\bar{b}/c\bar{c}$ followed by semileptonic decays have the potential to mimic the signal



(a) (b)

Figure 1: $m_{Z(\ell\ell)\ell'}$ mass distributions in the control regions; (a) for $N^{\pm}$ candidates in the ZZ control region and (b) for $N^{\pm}$ candidates in the Z+jets control region. The rightmost bin in each respective histogram includes overflow events.

signature. Although the event selection criteria are chosen to reject those Z+jets events, specifically the impact parameter and isolation requirements, the large cross section of this process means some events will pass the event selection. In the signal region, the expected number of Z+jets events, which includes $Z + b\bar{b}/c\bar{c}$, is approximately one. The uncertainty on this prediction is very large, due to the very large statistical uncertainties on the MC and data in the signal region and sideband, respectively. For this reason the Z+jets background shape is taken from MC, using a selection where the impact parameter requirement on the third lepton is reversed, its isolation requirements are loosened, and only three or more leptons are required in the event. Figure 1(b) shows that the MC shape is in good agreement with the data. The smaller $t\bar{t}V$, and VVV backgrounds evaluated directly from MC simulation have significant theoretical uncertainties due to poorly known cross sections for these processes.

4 Systematic Uncertainties

This section lists the sources of systematics uncertainties considered. The systematic uncertainties do not depend on the reconstructed heavy fermion mass. The difference between the fast simulation and the full Geant4 simulation gives a 6.8% systematic uncertainty on the signal acceptance. PDF uncertainties are evaluated, following Refs. [49, 28, 50, 51], using the envelopes of uncertainty sets as well as different PDF sets. Due to the limited statistical power of the Z+jets sample, a 100% systematic uncertainty is assigned to the yield in the loose three lepton region. The limited MC sample size of Z+jets in the signal region adds a normalisation uncertainty of 370%, corresponding to the 95% confidence level (CL) Poisson upper limit. The uncertainty associated to the VVV events is derived from Ref. [52].

The reconstruction, identification and trigger efficiencies for electrons and muons, as well as their momentum scales and resolutions, are estimated using $Z \to \ell\ell$, $J/\psi \to \ell\ell$, and $W \to \ell\nu$ decays ($\ell = e, \mu$). With the exception of the uncertainty on the electron selection efficiency and the $t\bar{t}$ sample, the resulting uncertainties are all smaller than 1%. Systematic uncertainties are assigned to the simulation scale factor. The uncertainty on the integrated luminosity is 3.6% [53]. All systematic uncertainties considered are listed in Table 3. For each source, the larger deviation from the central value was taken as the systematic uncertainty. The systematic uncertainties on the background yields are considered to be independent of $m_{Z(\ell\ell)\ell'}$. All systematic uncertainties are added in quadrature.

Table 3: Summary of the systematic uncertainties on the normalisation of the signal (m_N = 120 GeV) and background contributions, given in percentages.. The entries filled with a hyphen correspond to uncertainties that are consistent with zero or to systematic variations that are not applicable to the respective sample. The luminosity uncertainty is 3.6%, and is not included in this table.

	ZZ	Z+jets	VVV	$t\bar{t}V$	Signal [120 GeV]
E_e Resolution	0.2	–	< 0.1	< 0.1	0.3
E_e Scale	0.1	–	0.3	0.6	0.6
e Identification	2.7	2.8	2.8	2.7	2.7
μ Res. ID	0.1	–	< 0.1	1.7	0.1
μ Res. Spectr.	0.1	–	< 0.1	1.7	0.1
E_μ Scale	< 0.1	–	< 0.1	5.8	0.2
Shape	–	100	–	–	–
Scale Factor	–	370	–	–	–
Fast sim.	–	–	–	–	6.8
Signal PDF	–	–	–	–	0.9
Cross Section	6.4	11	100	50	–
Total	7.0	390	100	50	7.4

5 Results

This analysis is sensitive to events with at least four isolated leptons, arising from $pp \to N^\pm N^0$, with subsequent decays $N^\pm \to Z\ell^\pm$, $Z \to \ell^+\ell^-$ and $N^0 \to W^\pm\ell^\mp$. This decay mode dominates the contribution to this topology in the Type III Seesaw model. The total expected background in the signal region is 19.8 ± 0.2(stat.) ± 5.4(sys.) events. The distribution of $m_{Z(\ell\ell)\ell'}$ in the signal region is shown in Figure 2. The shape and normalisation are in agreement with the SM expectation.

Limits are then placed on the product of the cross section and branching fractions:

$$\sigma(pp \to N^\pm N^0)\mathcal{B}(N^\pm \to Z\ell^\pm)\mathcal{B}(N^0 \to W^\pm\ell^\mp) = \frac{N_{\text{evt}}}{A\varepsilon \int \mathcal{L}\mathrm{d}t} \tag{1}$$

where N_{evt} is the number of events, A is the acceptance, ε is the selection efficiency, which includes $\mathcal{B}(Z \to \ell^+\ell^-)$ in the mass window, and $\int \mathcal{L}\mathrm{d}t$ is the integrated luminosity of the data sample. Values of $A \times \varepsilon$ for different values of m_N are listed in Table 4.

The cross section times branching fraction limit is calculated as a function of the heavy fermion mass m_N, within mass windows. The optimal width of these windows depends on the mass of the heavy fermion under study, increasing in width for higher masses. Limits are determined at 16 mass points between 110 GeV to 260 GeV with ± 10 GeV mass windows to account for detector resolution, and 14 mass points between 280 GeV and 540 GeV with ±20 GeV mass windows. The number of expected and observed events after all selection requirements are given for a few representative mass windows in Table 5.

No significant excess of events is observed relative to the SM expectations. Therefore an upper limit at the 95% confidence level is set on $\sigma \times \mathcal{B}$, on the production cross section of N^0N^+ and its branching fraction, $\mathcal{B}(N^\pm \to Z\ell^\pm)\mathcal{B}(N^0 \to W^\pm\ell^\mp)$, to four lepton final states, via $N^\pm \to Z\ell^\pm$ and $N^0 \to W^\pm\ell^\mp$,

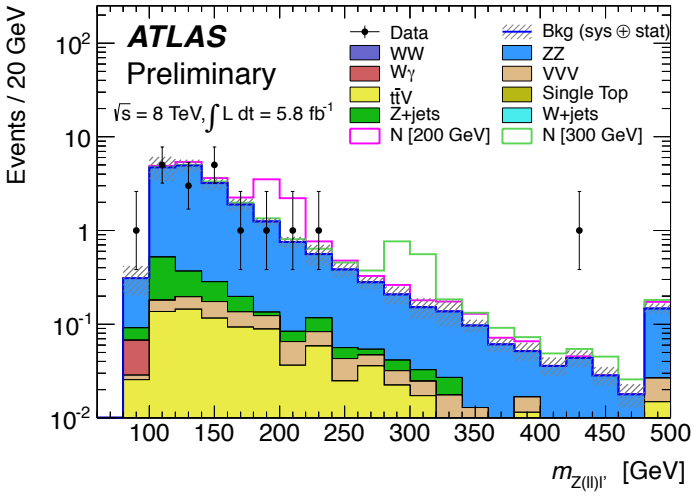


Figure 2: Invariant mass distribution of the $N^{\pm}$ candidates, $Z(\ell\ell)\ell'$, in the signal region, for data (black points), and the expected total background (solid histograms). The rightmost bins in the histograms include overflow events.

Table 4: The product of the signal acceptance and efficiency (including $\mathcal{B}(Z \to \ell\ell)$) for each simulated mass point.

m_N [GeV]	$A\varepsilon$
100	0.0006
120	0.0091
160	0.012
200	0.011
300	0.010
500	0.006

Table 5: The expected and observed event yields inside and outside of each mass bin, where the uncertainties include both statistical and systematic added in quadrature.

m_N	Expected Events (Inside signal mass bin)					Observed	Signal	Outside signal mass bin		
(GeV)	ZZ	Z+jets	VVV	$t\bar{t}V$	Total	Events	Events	Expected	Observed	Signal only
120	5.0 ± 0.4	< 1.7	0.05 ± 0.05	0.16 ± 0.08	5.7 ± 1.7	4	17.8 ± 1.2	14.5 ± 1.3	15	2.8 ± 0.5
160	2.2 ± 0.2	< 0.4	0.05 ± 0.05	0.10 ± 0.05	2.5 ± 0.5	3	7.6 ± 0.5	16.1 ± 0.2	16	2.9 ± 0.3
200	0.85 ± 0.07	<0.15	0.03 ± 0.03	0.05 ± 0.03	1.0 ± 0.2	1	1.0 ± 0.1	18.9 ± 1.3	18	0.6 ± 0.1
300	0.28 ± 0.03	<0.15	0.02 ± 0.02	0.04 ± 0.02	0.4 ± 0.2	0	0.25 ± 0.02	19.4 ± 1.3	19	0.19 ± 0.02
500	0.03 ± 0.01	<0.02	0.004 ± 0.004	0.006 ± 0.003	0.04 ± 0.04	0	0.015 ± 0.002	19.7 ± 1.3	19	0.059 ± 0.005

Table 6: The expected and observed limits on m_N obtained using the frequentist method, specified at a 95% confidence level, for six assumed sets of branching fractions.

	Mass limit [GeV]	
$\mathcal{B}(N^{\pm} \to Z\ell^{\pm})\mathcal{B}(N^0 \to W^{\pm}\ell^{\mp})$	Expected	Observed
1.00	350	350
0.66	326	330
0.33	284	290
0.25	266	280
0.15	240	230

where the lepton can be an electron or a muon. The branching fractions of $N^0 \to W^{\pm}\ell^{\mp}$ and $N^{\pm} \to Z\ell^{\pm}$ depend on m_N [19].

The frequentist CLs method is used to determine 95% CL upper limits on the number of events for each mass bin [54]. The upper limits on $\sigma\mathcal{B}(N^{\pm} \to Z\ell^{\pm})\mathcal{B}(N^0 \to W^{\pm}\ell^{\mp})$ are determined as a function of the fermion mass m_N, shown in Figure 3. Two theoretical cross section curves are overlaid, corresponding to maximal (unity) and the predicted values of the product of branching fractions, $\mathcal{B}(N^{\pm} \to Z\ell^{\pm})\mathcal{B}(N^0 \to W^{\pm}\ell^{\mp})$. The predicted value peaks at 0.20 at 135 GeV and asymptotically falls to 0.13 from 500 GeV onwards [19]. The mass limit on m_N is also determined as a function of the product of branching fractions $\mathcal{B}(N^{\pm} \to Z\ell^{\pm})\mathcal{B}(N^0 \to W^{\pm}\ell^{\mp})$, shown in Figure 4. In Table 6, lower limits on the mass are given for several values of $\mathcal{B}(N^{\pm} \to Z\ell^{\pm})\mathcal{B}(N^0 \to W^{\pm}\ell^{\mp})$.

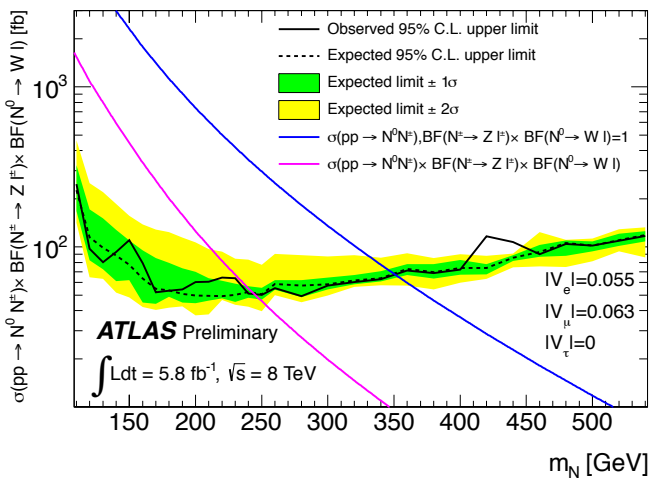


Figure 3: The expected (dashed line) and observed (solid line) exclusion limits at 95% confidence level on $\sigma\mathcal{B}$ as a function of the fermion mass m_N assuming $|V_e| = 0.055$, $|V_\mu| = 0.063$ and $|V_\tau| = 0$ and for $\mathcal{B}$=1 (theory LO). The dark(green) and light (yellow) shaded areas represent the 1 standard deviation (68% C.L.) and 2 standard deviations (95% CL) limits on the expected, respectively. At $m_N = 420$ GeV, the probability to have equal to or more than the observed number of events with a background only hypothesis, p_0, is found to be 0.20.

Assuming $\mathcal{B}(N^{\pm} \to Z\ell^{\pm})\mathcal{B}(N^0 \to W^{\pm}\ell^{\mp}) = 1$, a limit of $m_N > 350$ GeV is expected: the observed limit is $m_N > 350$ GeV. When $\mathcal{B}(N^{\pm} \to Z\ell^{\pm})\mathcal{B}(N^0 \to W^{\pm}\ell^{\mp})$ is set to the nominal mass dependent

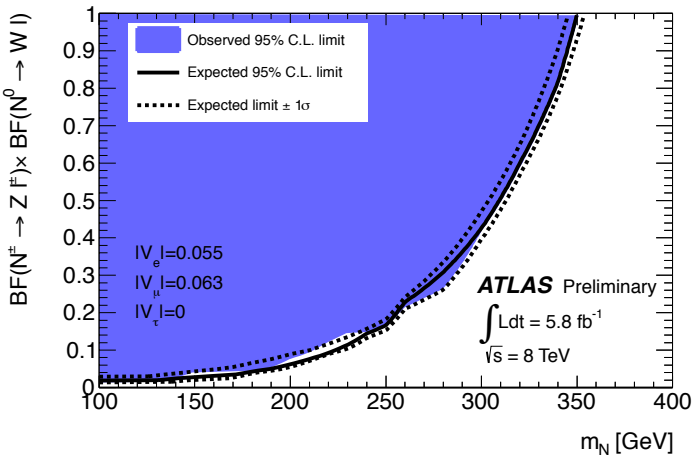


Figure 4: The expected (dashed line) and observed (shaded region) upper limits at 95% confidence level on $\mathcal{B}(N^{\pm} \to Z\ell^{\pm})\mathcal{B}(N^{0} \to W^{\pm}\ell^{\mp})$ versus m_N assuming $|V_e| = 0.055$, $|V_\mu| = 0.063$ and $|V_\tau| = 0$.

values, the expected limit is reduced to $m_N > 243$ GeV, and the observed limit becomes $m_N > 245$ GeV. At $m_N = 420$ GeV, the probability to have equal to or more than the observed number of events with a background only hypothesis, p_0, is found to be 0.20.

This analysis inclusively searches for heavy fermions decaying into electron and muon final states. As the electron and muon reconstruction efficiencies are similar, the limit therefore holds true for cases where the electron and muon couplings are similar in size. In the case where the tau coupling is non-negligible, the mass limit from this measurement will be reduced.

6 Conclusions

A search for evidence of Type III Seesaw heavy fermions, manifesting as a trilepton resonance, was performed using events containing four isolated leptons (muons or electrons) with a reconstructed Z boson. The dataset used is from pp collisions at $\sqrt{s} = 8$ TeV corresponding to an integrated luminosity of 5.8 fb^{-1}, recorded during 2012 by the ATLAS experiment at the LHC. No evidence for pair production of $N^{\pm}N^{0}$ states has been found, and 95% confidence upper limits are set on the production cross section of $N^{\pm}N^{0}$ times the branching fractions of the neutral heavy fermion to $W^{\pm}\ell^{\mp}$ and of the charged heavy fermion to $Z\ell^{\pm}$. Assuming the Seesaw type III model prediction for the product of branching fractions, $\mathcal{B}(N^{\pm} \to Z\ell^{\pm})\mathcal{B}(N^{0} \to W^{\pm}\ell^{\mp})$ equal to the mass dependent predicted values, lower bounds on the mass of the N states are placed at 245 GeV at 95% CL.

References

[1] Super-Kamiokande Collaboration, Y. Fukuda et al., Phys.Rev.Lett. **81** (1998) 1562–1567, `arXiv:hep-ex/9807003 [hep-ex]`.

[2] D. V. Forero, M. Tortola and J. W. F. Valle, Phys.Rev. **D86** (2012) 073012, `arXiv:1205.4018 [hep-ph]`.

[3] A. Goobar, S. Hannestad, E. Mortsell and H. Tu, JCAP **0606** (2006) 019, `arXiv:astro-ph/0602155 [astro-ph]`.

[4] Y. Chikashige, R. N. Mohapatra, and R. D. Peccei, Phys.Lett. **B98** (1981) 265.

[5] P. Minkowski, Physics Letters B **67** (1977) 421 – 428.

[6] T. Cheng and L.-F. Li, Phys. Rev. D **22** (1980).

[7] G. B. Gelmini and M. Roncadelli, Phys.Lett. **99B** (1981).

[8] T. Han and B. Zhang, Phys.Rev.Lett. **97** (2006) 171804, `arXiv:hep-ph/0604064 [hep-ph]`.

[9] F. del Aguila,J. Aguilar-Saavedra, and R. Pittau, JHEP **0710** (2007) 047, `arXiv:hep-ph/0703261 [hep-ph]`.

[10] J. Kersten and A. Y. Smirnov, Phys.Rev. **D76** (2007) 073005, `arXiv:0705.3221 [hep-ph]`.

[11] M. Muhlleitner and M. Spira, Phys.Rev. **D68** (2003) 117701, `arXiv:hep-ph/0305288 [hep-ph]`.

[12] A. Akeroyd and M. Aoki, Phys.Rev. **D72** (2005) 035011, `arXiv:hep-ph/0506176 [hep-ph]`.

[13] A. Hektor,M. Kadastik,M. Muntel,M. Raidal, and L. Rebane, Nucl.Phys. **B787** (2007) 198–210, `arXiv:0705.1495 [hep-ph]`.

[14] P. Fileviez Perez,T. Han,G.-Y. Huang,T. Li, and K. Wang, Phys.Rev. **D78** (2008) 071301, `arXiv:0803.3450 [hep-ph]`.

[15] W. Chao,Z.-G. Si,Z.-z. Xing, and S. Zhou, Phys.Lett. **B666** (2008) 451–454, `arXiv:0804.1265 [hep-ph]`.

[16] R. Foot, H. Lew and X.-G. He and G.C. Joshi, Z. Phys. **C 44** (1989) 441.

[17] R. Franceschini, T. Hambye, and A. Strumia, Phys.Rev. **D78** (2008) 033002, `arXiv:0805.1613 [hep-ph]`.

[18] F. del Aguila and J. Aguilar-Saavedra, Nucl.Phys. **B813** (2009) 22–90, `arXiv:0808.2468 [hep-ph]`.

[19] C. Biggio and F. Bonnet, Eur.Phys.J. **C72** (2012) 1899, `arXiv:1107.3463 [hep-ph]`.

[20] F. del Aguila, J. de Blas, M. Perez-Victoria, Phys.Rev. **D78** (2008) 013010, `arXiv:0803.4008 [hep-ph]`.

[21] A. Abada, C. Biggio, F. Bonnet, M. B. Gavela and T. Hambye, Phys.Rev. **D78** (2008) 033007, `arXiv:0803.0481 [hep-ph]`.

[22] A. Abada, C. Biggio, F. Bonnet, M. B. Gavela and T. Hambye, JHEP **0712** (2007) 061, `arXiv:0707.4058 [hep-ph]`.

[23] CMS Collaboration, Phys.Lett. **B718** (2012) 348–368, `arXiv:1210.1797 [hep-ex]`.

[24] ATLAS Collaboration, JINST **3** (2008) S08003.

[25] T. Sjostrand, S. Mrenna and P. Z. Skands, JHEP **0605** (2006) 026, `arXiv:hep-ph/0603175 [hep-ph]`.

[26] T. Sjostrand, S. Mrenna and P. Z. Skands, Comput.Phys.Commun. **178** (2008) 852–867, `arXiv:0710.3820 [hep-ph]`.

[27] G. Corcella et al., JHEP **0101** (2001) 010, arXiv:hep-ph/0011363 [hep-ph].

[28] H.-L. Lai et al., Phys.Rev. **D82** (2010) 074024, arXiv:1007.2241 [hep-ph].

[29] M. Mangano et al., JHEP **0307** (2003) 001.

[30] J. Alwall et al., Eur.Phys.J. **C53** (2008) 473, arXiv:0706.2569 [hep-ph].

[31] J. Alwall, M. Herquet, F. Maltoni, O. Mattelaer and T. Stelzer, JHEP **1106** (2011) 128, arXiv:1106.0522 [hep-ph].

[32] J. M. Campbell and R. K. Ellis, Phys.Rev. **D60** (1999) 113006, arXiv:hep-ph/9905386 [hep-ph].

[33] ATLAS Collaboration, ATLAS-CONF-2012-098, https://cds.cern.ch/record/1462530/.

[34] S. Frixione and B. R. Webber, JHEP **06** (2002) 029.

[35] B. P. Kersevan and R. W. Elzbieta, arXiv:hep-ph/0405247.

[36] T. Binoth, N. Kauer and P. Mertsch, arXiv:0807.0024 [hep-ph].

[37] T. Binoth, M. Ciccolini, N. Kauer and M. Kramer, JHEP **12** (2006) 046.

[38] S. Alioli, P. Nason, C. Oleari and E. Re, JHEP **0904** (2009) 002.

[39] P. Nason and C. Oleari, JHEP **1002** (2010) 037.

[40] P. M. Nadolsky et al., Phys.Rev. **D78** (2008) 013004, arXiv:0802.0007 [hep-ph].

[41] N. L. Christensen and C. Duhr, Comput. Phys. Commun. **180** (2009) 1614.

[42] P. Meade and M. Reece, arXiv:hep-ph/0703031 [hep-ph].

[43] J. Alwall, P. Demin, S. de Visscher, R. Frederix, M. Herquet and others, JHEP **0709** (2007) 028, arXiv:0706.2334 [hep-ph].

[44] ATLAS Collaboration, Eur.Phys.J. **C70** (2010) 823, arXiv:1005.4568 [physics.ins-det].

[45] GEANT4 Collaboration, S. Agostinelli et al., Nucl. Instrum. Meth. **A506** (2003) 250–303.

[46] ATLAS Collaboration, Eur. Phys. J. **C 70** (2010) 823.

[47] ATLAS Collaboration, Eur.Phys.J. **C72** (2012) 1909, arXiv:1110.3174 [hep-ex].

[48] ATLAS, ATLAS-CONF-2011-063, https://cds.cern.ch/record/1345743/.

[49] A. D. Martin, W. J. Stirling, R. S. Thorne and G. Watt, Eur.Phys.J. **C63** (2009) 189, arXiv:0901.0002 [hep-ph].

[50] R. D. Ball et al., Nucl.Phys. **B849** (2011) 296, arXiv:1101.1300 [hep-ph].

[51] M. Botje et al., arXiv:1101.0538 [hep-ph].

[52] ATLAS Collaboration, ATLAS-CONF-2012-163, https://cds.cern.ch/record/1494191/.

[53] ATLAS Collaboration, Eur.Phys.J. **C71** (2011) 1630, arXiv:1101.2185 [hep-ex].

[54] A. L. Read, J. Phys. **G28** (2002) 2693.